Lab-04 Data Science Hands-on Guide

Overview

This is a hands-on lab that demonstrates the end-to-end ML pipeline for forecasting daily stock inventory levels across multiple SKUs using Azure Databricks. Followed by how to invoke the **inventory-forecast-endpoint** using an AI agent connected via the **Model Context Protocol (MCP).**

The goal is to provide a hands-on experience in creating an end-to-end working agentic solution using a custom created ML model in Azure Databricks.

You will complete the following steps during the session.

Step	Description
1	**Configuration** — Set experiment directory and model names
2	**Synthetic Data** — Generate daily inventory for 5 SKUs with trend, seasonality & replenishment
3	**EDA** — Visualise time series patterns, distribution and seasonal effects
4	**Databricks AutoML** — Auto-train & tune forecasting models (10-minute budget)
5	**Prediction** — Load best model, generate 30-day forward forecasts
6	**Install packages** — `databricks-langchain`, `mcp`, `fastmcp`
7	**Configuration** — Endpoint name, workspace URL, LLM endpoint
8	**Define MCP Server** — FastMCP server exposing the forecast endpoint as a tool
9	**Build Agent** — LangGraph ReAct agent connected to the MCP server
10	**Test Agent** — Invoke forecasts via natural language queries

Prerequisite

Clone Repo

Please clone the following repo: git clone https://github.com/mufajjul/ai-workshop-infra.git

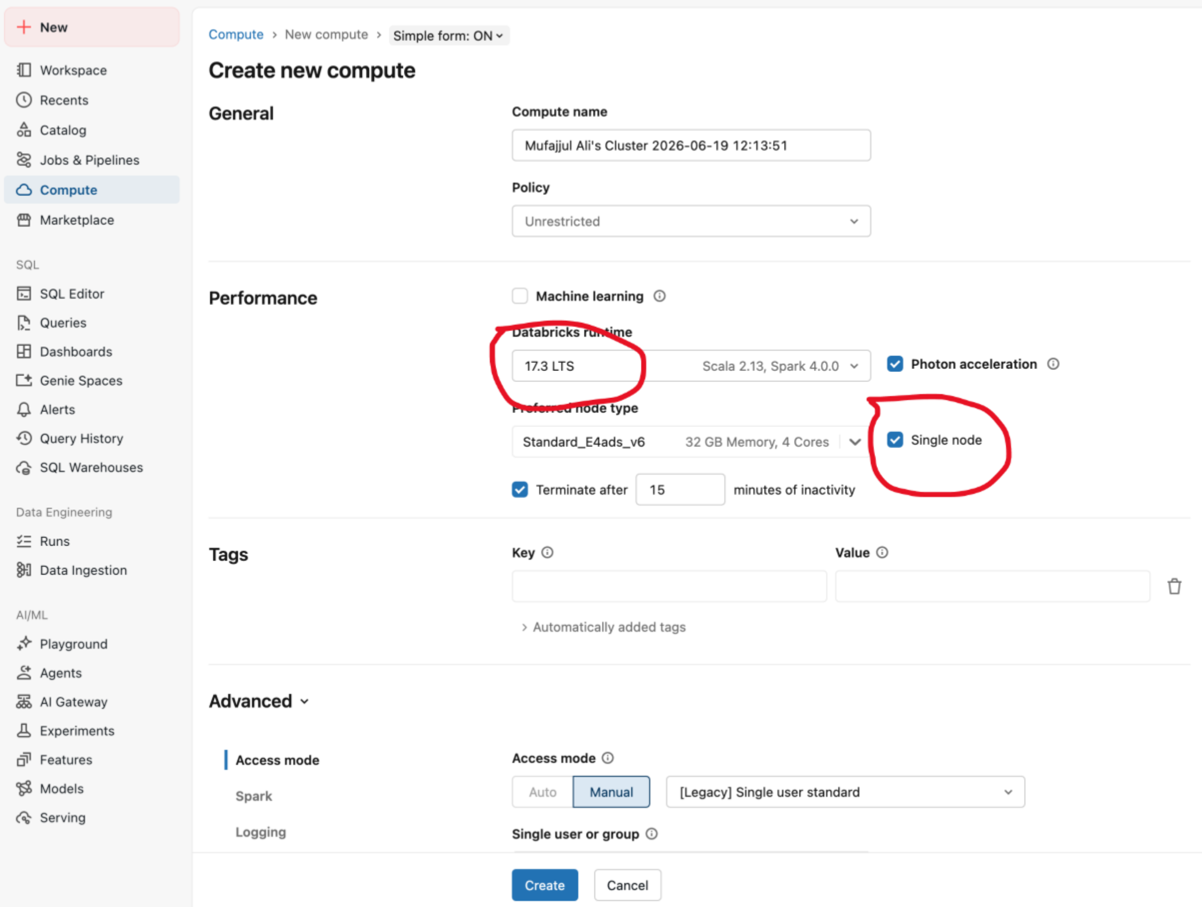
Environment provisioning.

Please run the following script to provision the DB workspace.

```
sh -l infra/Databricks/ARM/cli.sh
```

Activity 1: Cluster Creation

Please create a cluster per person for this session. Recommendation is to use a **single node cluster** to avoid any sub-level SKU quota issue.



Databricks runtime: Make sure you select the Databricks runtime to 17.3 LTS. AutoML is not available for high versions.

Cluster: The cluster is selected as “single node”.

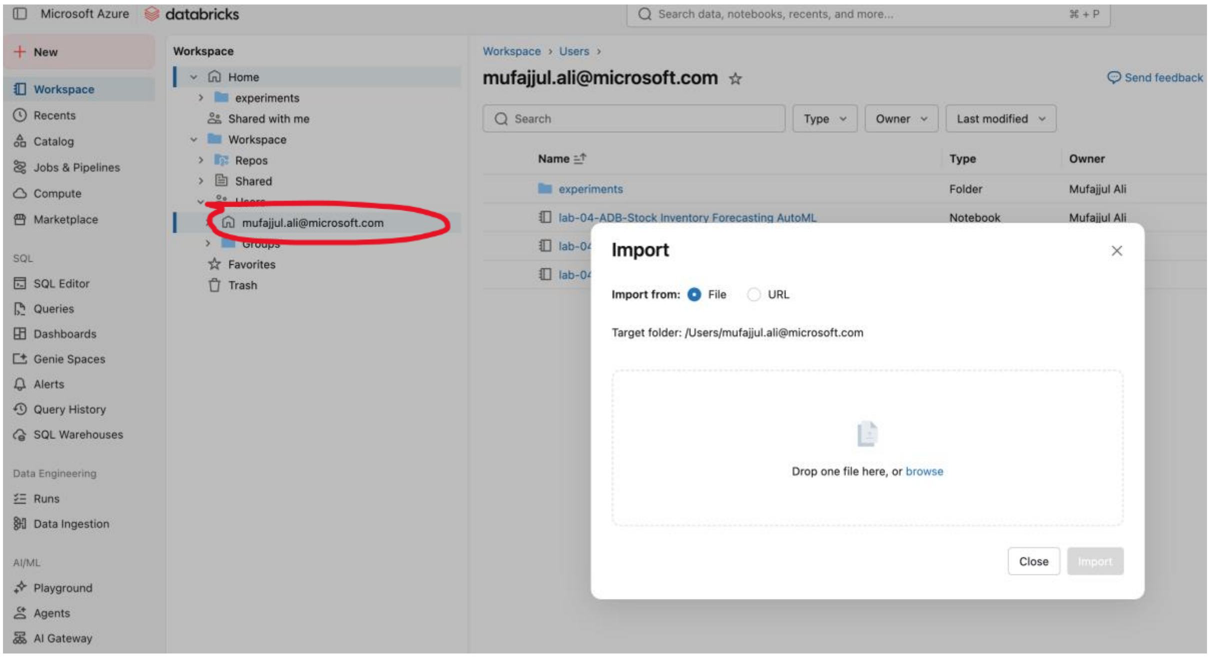
SKU: Select Standard_E4ads_V6 (recommended), you can also use other types as well, especially if quota is not available.

Activity 2: Load the notebooks

There are two notebooks used for this lab:

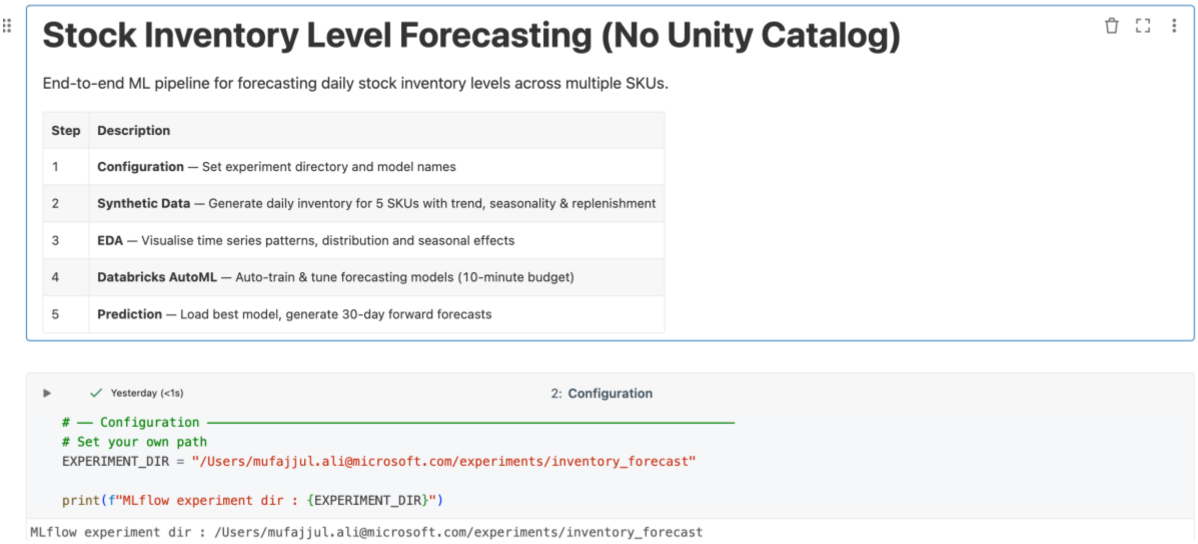
- data-science/session3-lab4-adb/lab-04-ADB-Stock Inventory Forecasting AutoML (No UC).ipynb
- data-science/session3-lab4-adb/lab-04a-Inventory Forecast Agent with MCP.ipynb

Both labs, need to be imported into the workspace. Please right click on your workspace and select import.



Activity 3: Run notebook 1

Pease open the notebook, lab-04-ADB-Stock Inventory Forecasting AutoML (No UC).ipynb, and run individual cells.



Stock Inventory Level Forecasting (No Unity Catalog)

End-to-end ML pipeline for forecasting daily stock inventory levels across multiple SKUs.

Step	Description
1	**Configuration** — Set experiment directory and model names
2	**Synthetic Data** — Generate daily inventory for 5 SKUs with trend, seasonality & replenishment
3	**EDA** — Visualise time series patterns, distribution and seasonal effects
4	**Databricks AutoML** — Auto-train & tune forecasting models (10-minute budget)
5	**Prediction** — Load best model, generate 30-day forward forecasts

Yesterday (<1s) 2: Configuration

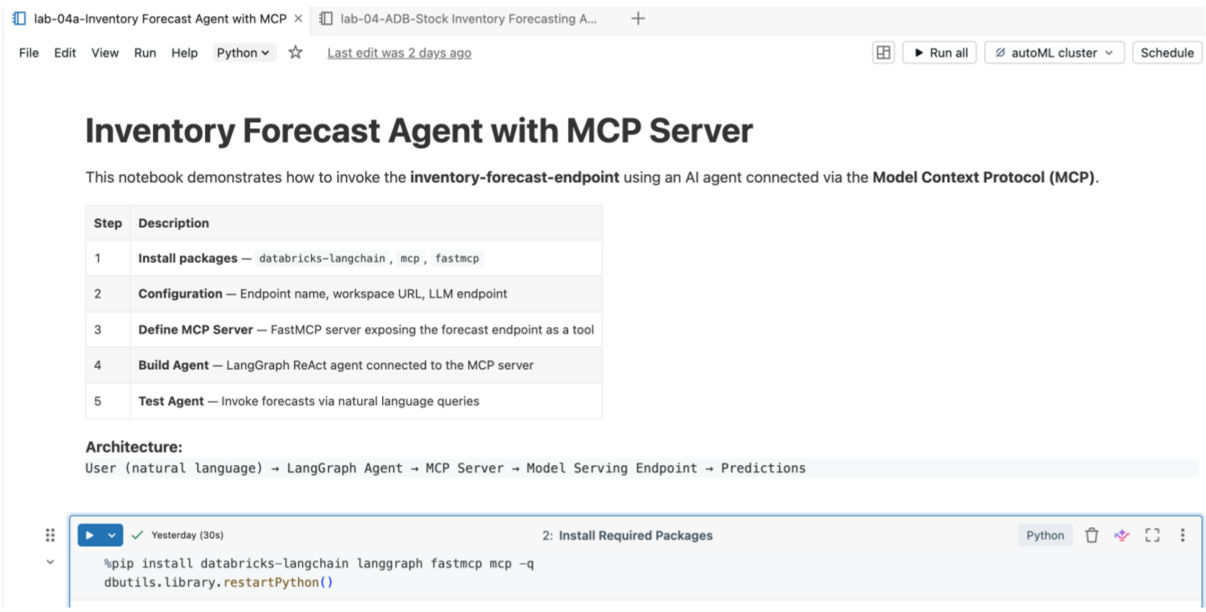
```
# — Configuration ————————————————————————————————
# Set your own path
EXPERIMENT_DIR = "/Users/mufajjul.ali@microsoft.com/experiments/inventory_forecast"

print(f"MLflow experiment dir : {EXPERIMENT_DIR}")
```

```
MLflow experiment dir : /Users/mufajjul.ali@microsoft.com/experiments/inventory_forecast
```

There is a total of six tasks to be completed here. Please go through each one of them, once you have completed, there should be an endpoint deployed with the best model.

Activity 4: Run notebook 2

Pease open the notebook, lab-04-Inventory Forecast Agent with MCP.ipynb, and run individual cells.



lab-04a-Inventory Forecast Agent with MCP × lab-04-ADB-Stock Inventory Forecasting A... +

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Inventory Forecast Agent with MCP Server

This notebook demonstrates how to invoke the **inventory-forecast-endpoint** using an AI agent connected via the **Model Context Protocol (MCP)**.

Step	Description
1	**Install packages** — `databricks-langchain`, `mcp`, `fastmcp`
2	**Configuration** — Endpoint name, workspace URL, LLM endpoint
3	**Define MCP Server** — FastMCP server exposing the forecast endpoint as a tool
4	**Build Agent** — LangGraph ReAct agent connected to the MCP server
5	**Test Agent** — Invoke forecasts via natural language queries

Architecture:

`User (natural language) → LangGraph Agent → MCP Server → Model Serving Endpoint → Predictions`

Yesterday (30s) 2: Install Required Packages Python

```
%pip install databricks-langchain langgraph fastmcp mcp -q
dbutils.library.restartPython()
```

There are five tasks to be completed here, please go through each one of them, you should be able to test an agent to use the custom model to perform various asks.

Note: The deployed model can also be invoked/used by the agents created in Azure Foundry.